

Poster Presentation Rubric

EEB 187: Ecology and Evolution of Color | Spring 2026

Group:	Clade:
Date:	Grader:

Presentations: Tue Jun 3 (Group A) / Thu Jun 5 (Group B)

Scoring: 8 categories, each on a 4-point scale. Total: **32 points**.

- **4 = Excellent** – exceeds expectations, ready for a conference
- **3 = Good** – meets expectations with minor issues
- **2 = Adequate** – meets minimum requirements but with notable gaps
- **1 = Needs Work** – does not meet expectations; major revisions needed

Part A: Poster Design and Physical Elements (50%, 16 points)

Category 1: Visual Clarity and Layout

Score	Descriptor
4	Poster follows the 3-column UCLA template layout cleanly. Title is readable from 6 feet. Section headers (36–48 pt) and body text (24–28 pt) follow the font hierarchy. Figures are large and well-positioned. White space is used effectively – the poster does not feel crowded or empty. The eye flows naturally from Introduction through Results to Conclusions.
3	Layout is organized and mostly readable. One minor issue: slightly small captions, one crowded section, or a figure positioned awkwardly. The overall flow is still clear.
2	Layout has noticeable problems: text is too small to read at 3 feet, figures are crammed together, or white space is distributed unevenly. The reading path is unclear.
1	Layout is disorganized. Text is unreadable at conversational distance. No clear visual hierarchy. Sections are missing labels or placed out of order.

Category 2: Figure Quality

Score	Descriptor
4	All hero figures are poster-ready: high resolution (300 dpi), labeled axes with variable names and units (e.g., “PC1 – chromatic boundary strength, 39.5% var.”), appropriate color palettes, no default R gray backgrounds, and informative captions (italic, 18–20 pt) that explain what each figure shows without needing body text.
3	Figures are clear and high resolution. One minor issue: a missing axis unit, a caption that is too brief, or one figure slightly pixelated.
2	Figures have multiple problems: blurry or low-resolution images, unlabeled axes, default R styling (gray backgrounds, tiny axis text), or missing captions.
1	Figures are unusable: screenshot-quality resolution, no axis labels, no captions, or figures are clearly placeholders.

Category 3: Scientific Content and Completeness

Score	Descriptor
4	All required sections are present and complete: Introduction (background on clade + why color matters), Research Question (single clear sentence), Hypothesis (testable predictions with expected outcomes), Methods (pavo pipeline, tree source, PCM approach), Results (figures with captions), Discussion (interpretation in context), Conclusions (key takeaways), References (3–5 citations in APA), Acknowledgments. The narrative is coherent – each section connects logically to the next.
3	All sections present. The narrative is mostly coherent but one section is underdeveloped (e.g., Discussion is vague, Methods lacks detail on the tree source, or Hypothesis restates the question instead of making a prediction).
2	One or more sections are missing or severely underdeveloped. The narrative has gaps – the reader cannot follow the logic from question to conclusion without guessing.
1	Multiple sections missing. The poster reads as a disconnected collection of figures and text fragments. No clear research question or hypothesis.

Category 4: Analytical Rigor

Score	Descriptor
4	The group correctly ran the required recipes (color gamut, phylogenetic signal, axis of diversification) plus 1–2 electives. Statistical results are accurately reported with appropriate notation (e.g., $K = 0.32$, $p = 0.001$; PGLS slope = 1.47, SE = 0.52, $p = 0.006$; lambda = 0.64). Interpretations are valid – the group distinguishes significant from non-significant results, interprets K relative to Brownian motion expectations, and does not overstate conclusions. Non-significant results are reported honestly and interpreted constructively.
3	Required analyses were run and results are mostly reported correctly. One minor issue: a missing p-value, lambda not reported for PGLS, or a slight overstatement (“we proved that...” instead of “our results are consistent with...”).
2	Analyses were run but results are reported incompletely or with errors. K is misinterpreted (e.g., “ $K = 0.3$ means no phylogenetic signal”), p-values are missing, or the group conflates correlation with causation.

Score	Descriptor
1	Required analyses are missing or clearly incorrect. Results are fabricated or copied from another group. The group cannot explain what their statistical output means.

Part B: Oral Presentation (50%, 16 points)

These categories assess whether presenters can explain the analytical arc reinforced in Thursday's Lecture 18 synthesis. Each presenter should be prepared to address any of these topics.

Category 5: Explain the Color Gamut

Score	Descriptor
4	The presenter clearly explains what their clade's CIELAB gamut shows: which colors are present, which are absent, and why. They correctly describe the a* (green–red) and b* (blue–yellow) axes as opponent-process dimensions (not wavelengths). They can state the gamut volume as a percentage of sRGB and explain what it means biologically (e.g., “our clade uses 18% of visible color space, dominated by yellows and browns, with almost no reds or greens”).
3	The presenter describes the gamut accurately but with one gap: cannot explain what a* and b* represent, or states the gamut volume without interpreting it biologically.
2	The presenter can point to the gamut figure but cannot explain the axes or what the shape of the cloud means. Relies on reading the poster text aloud rather than explaining in their own words.
1	The presenter cannot explain what the gamut figure shows or confuses CIELAB axes with wavelength.

Category 6: Interpret the Morphospace and Phylogeny

Score	Descriptor
4	The presenter explains what PC1 and PC2 capture by referencing the top loading variables (e.g., “PC1 is driven by transition density and boundary strength – it separates complex-patterned species from plain ones”). They explain what Blomberg's K means for their clade in plain English (e.g., “K = 0.3 means color is more labile than Brownian motion predicts – close relatives don't look more similar than expected”). They can walk through the ASR tree and identify where major color transitions occurred.
3	The presenter explains PC axes and K correctly but with one gap: cannot name the top loading variables, or states K without comparing to the Brownian expectation of 1.
2	The presenter can identify that PCA was used but cannot explain what the axes represent. States K as a number without interpretation. Cannot read the ASR tree.
1	The presenter cannot explain the morphospace or phylogenetic signal analysis. Confuses PCA axes with raw variables, or states “K is the phylogenetic signal” without further explanation.

Category 7: Motivate and Interpret the Evolutionary Test

Score	Descriptor
4	The presenter explains why PGLS is needed (species are not independent data points because they share evolutionary history – the Felsenstein problem). They state their hypothesis clearly, identify the predictor and response variables, and interpret the PGLS result biologically (e.g., “trophic level does not significantly predict PC1 after controlling for phylogeny, $p = 0.41$, which means diet alone does not explain pattern variation in our clade”). They can explain what lambda means in the context of their PGLS.
3	The presenter explains the non-independence problem and interprets the PGLS result correctly, but with one gap: does not mention lambda, or cannot explain what the slope coefficient means biologically.
2	The presenter can state the p-value but cannot explain why phylogenetic correction was needed. Interpretation is superficial (“the result was significant / not significant”) without biological meaning.
1	The presenter cannot explain why PGLS was used instead of ordinary regression, or misinterprets the result (e.g., “ $p > 0.05$ means we failed” or “PGLS controls for body size”).

Category 8: Respond to Questions and Make Connections

Score	Descriptor
4	The presenter handles follow-up questions confidently, connecting their clade’s results to course concepts from multiple weeks (e.g., “the yellow-dominated gamut is consistent with carotenoid pigments from Lecture 8, and the low K we found matches the Turing model prediction from Lecture 13 that pattern switching requires few genetic changes”). They can compare their clade’s results to the instructor’s Chaetodontidae example or to another group’s clade. They acknowledge limitations honestly.
3	The presenter answers most questions correctly and makes at least one cross-topic connection. Occasionally hesitates or gives a partial answer but recovers with prompting.
2	The presenter can answer basic factual questions about their poster but cannot connect results to broader course themes. Struggles with “why” and “how does this relate to...” questions.
1	The presenter cannot answer questions beyond what is written on the poster. No connections to course material. Defers all questions to other group members.

Quick Score Sheet

Circle one score per category. Sum for total.

Group:	Clade:
Date:	Grader:

PART A: POSTER DESIGN (16 pts)

#	Category	4	3	2	1	Score
1	Visual clarity and layout	4	3	2	1	_____
2	Figure quality	4	3	2	1	_____
3	Scientific content and completeness	4	3	2	1	_____
4	Analytical rigor	4	3	2	1	_____
Subtotal A:						_____ / 16

PART B: ORAL PRESENTATION (16 pts)

#	Category	4	3	2	1	Score
5	Explain the color gamut	4	3	2	1	_____
6	Interpret morphospace and phylogeny	4	3	2	1	_____
7	Motivate and interpret the evolutionary test	4	3	2	1	_____
8	Respond to questions and make connections	4	3	2	1	_____
Subtotal B:						_____ / 16

TOTAL: _____ / 32	Percentage: _____ %
-------------------	---------------------

Comments:
